Project Executable Files

Date	28 June 2026
Team ID	********
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for the OptiCrop project. All files are organized as shown below and uploaded to the SkillWallet submission platform. The evaluator can run the Flask application locally using the steps provided.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Overview of the OptiCrop project's file and folder structure:

```
OptiCrop/
├── app.py                        - Flask application (routes, prediction logic)
├── model.py                      - Model training script
├── analysis.ipynb                - Data analysis & EDA notebook (Epics 1-4)
├── Crop_recommendation.csv       - Dataset
├── Crop_Recommendation_Model.pkl - Saved trained ML model
```

```
├── requirements.txt           - Python dependencies
├── templates/                 - HTML pages (Jinja2 templates)
│   ├── index.html
│   ├── result.html
│   └── base.html
├── static/                    - CSS, JS, images for the web app
│   ├── css/
│   └── js/
└── README.md                  - Setup and run instructions
```

Step 3: Deployment / Access Details

Hosted / Deployed URL	Not deployed (runs locally via Flask development server)
Login Credentials (Demo)	Not applicable — no authentication required
Platform / Hosting Provider	Local / Flask dev server (update if deployed on Render, Heroku, etc.)
Repository Link	[Add your GitHub repository link here]
Demo Video Link	[Add your demo video link here]

Step 4: Run Instructions

Step-by-step instructions for the evaluator to run OptiCrop locally:

Pre-requisites:
Python 3.9+, pip, Anaconda (optional), Flask, scikit-learn, pandas, numpy, joblib

Step 1: Clone/extract the project folder: git clone <repository-url> (or unzip the submitted source code)
Step 2: Navigate into the project folder: cd OptiCrop
Step 3: Install dependencies: pip install -r requirements.txt
Step 4: (If model not already saved) Run model.py or analysis.ipynb to train and save Crop_Recommendation_Model.pkl
Step 5: Start the application: python app.py
Step 6: Open browser at: http://127.0.0.1:5000

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Model accuracy may vary slightly on retraining due to random_state variations	Fixed random_state=0 used for reproducibility
2	SMOTE / Jupyter re-run errors if cells executed out of order	Always run the imports cell first, then run cells sequentially
3	Application not yet deployed to a public host	Currently runs locally; deployment planned (Render/Heroku)